Pinch

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What is it?

- Personal baselines → advice fits *you*, not an average user
- One actionable nudge → clear $ impact, not a noisy dashboard
- Trust by design → transparent “Why?”, and P50/P80 goal dates

How it thinks

Signals we compute per category:

- **EMA baseline:** learns *your* normal
  - $\mathrm{EMA}_t = \alpha S_t + (1-\alpha)\mathrm{EMA}_{t-1}$, $\alpha \approx 0.3$
- **CPI-adjusted forecast:**
  - $\mathrm{Forecast}_t = \mathrm{EMA}_t \times (1 + \mathrm{CPI}_{\mathrm{yoy}})$
- **Variance (z-score):**
  - $z = \dfrac{\mathrm{actual} - \mathrm{forecast}}{\sigma_{\mathrm{robust}}}$ (MAD-based fallback when data is sparse)
- **Recurrence coverage:** fraction of spend from merchants seen in ≥N of last M months (down-weights predictable bills)

Selection Policy

Candidate actions we consider:

- Lower/weekly cap on volatile presets (e.g., transport, groceries)
- Route surplus to top goal (e.g., Emergency Fund)
- (Optional) Nudge to shift small habits (e.g., coffee → home brewing)

Scoring & choice:

- **Impact estimate:** monthly $ saved if action is applied
- **Score:** $\mathrm{impact}_i \times f(z_i) \times (1 - w \cdot \mathrm{recurrence}_i)$
- **ε-greedy choice:** exploit best score; with small ε (≈0.15), explore next-best

Output (the card):

- Title (e.g., “Variance in transport”), **$ impact**, Suggested action, **Why:** EMA / z / recurrence values

What Sets Pinch apart

- **Single decision, not a feed** → reduces choice overload
- **Cadence-aware** → recurring bills don’t cry wolf
- **Explainable “Why?”** → inputs shown on every card
- **Honest uncertainty** → **P50/P80** goal dates, not fake precision
- **Lightweight & offline-friendly** → CSV ingest, no bank tokens required for demo
- **Privacy-respecting** → minimal PII; transparent math

Why It Works

- **Clarity:** exactly one next action + concrete $ estimate
- **Personal relevance:** baselines are *yours* → fewer false positives
- **Low friction:** apply cap / route savings in one step
- **Trust:** anomalies dialed down for predictable cadence; uncertainty surfaced with P50/P80
- **Repeatable win:** one move per month → habit compounding

The Stack

- **Backend:** Python, FastAPI, Uvicorn, SQLAlchemy + SQLite
- **Modeling:** NumPy; EMA + CPI uplift + robust z-score + recurrence coverage
- **Policy:** small ε-greedy selector on $ impact
- **Auth:** JWT (HS256 / `JWT_SECRET`)
- **Frontends:** terminal runbook (curl + jq) and single-file HTML (Live/Offline)

Ethics

- **Explainability:** show EMA, z, recurrence used for each insight
- **Uncertainty:** P50/P80 goal dates
- **Minimize data:** merchant alias + totals, no bank tokens in MVP
- **Human-in-the-loop:** user must apply suggestions; no auto-actions
- **Known limits:** sparse data widens bands; recurrence improves with history